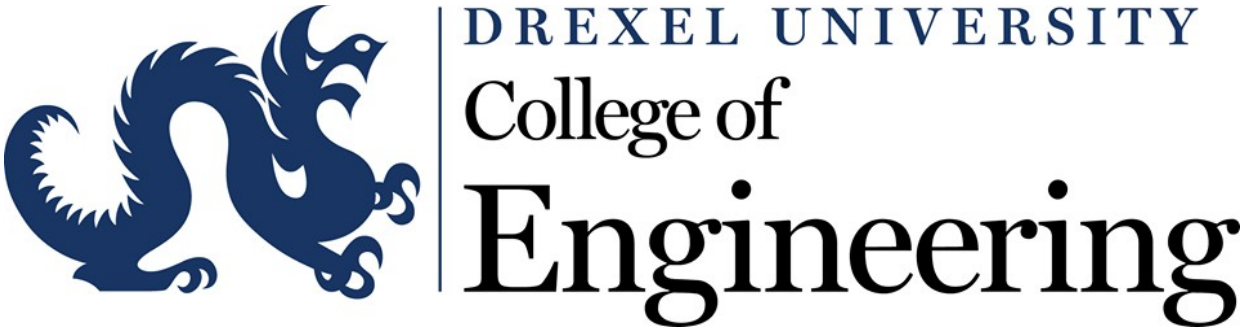


MEM-493 Senior Design Final Report

Project Title: Automated River Cleaner

Senior Design Group 32

Jie Xu
Xiao Tan
Tongming Huang
Yu Zheng
Chun Leung Devon Liu

Instructor:
Dr. B.C.Chang
Dr. Ahmad Najafi

Sponsor:
Hua Yuan Construction LLC.

Due Date: 5/13/2019

Executive Summary

HuaYuan Construction.LLC is facing an urgent problem. The government is planning to place a sanction because the employees of the companies throw too much trash into the river nearby the factory. After the negotiation with the local government, the company has to be responsible for the river cleaning and collect all the trash in the river. This will require an extensive amount of time and be very expensive if the company were to hire people to pick up all the garbage. Thus, the company is considering to use a robot to complete this task. Our group members are interested in solving this problem, and we had contact with the company. the company's owner is willing to be our sponsor and will fund us to design and build a river cleaning robot.

After communication with HuaYuan Construction.LLC, the stakeholder has following needs:

1. The robot can be charged by solar and wind energy.
2. The robot can detect ships, living animals and avoid collisions.
3. The robot is reliable and only needs minimal daily maintenance.
4. The robot can use GPS as a guide to patrol in the assigned working area.
5. The robot is easy to control and does not require more than one person to operate.

Our desired schedule for the production of the robot will mainly be separated into three stages. During the fall term, our group is going to design and build the boat platform. For the winter term, the devices and the energy system are going to be set up on the platform. In the spring term, the robot's functions are going to be programmed and tested. The product will come out in the last stage after testing.

Our final outcome is desired to be a robot that collects garbages automatically and with high energy efficiency.In addition to these goals, the robot be able to operate with little to no human operation and be able to avoid collisions.

Our fund is $15,000 and our group has a clear cost recorded document. Devon will record the cost and store the cost document in the one drive so that each member of the group can access the file.

Table of Contents:

I. Introduction **4**

1.1 Background 4
1.1.1 Updated Background 5
1.2 The Stakeholder's Needs and Specifications 5
1.2.1 Updated Stakeholder's Needs and Specifications 5
1.3 Problem Statement 6

II. Methods 6

III. Design Description 6
3.1 Design Idea Assumption 6
3.1.1 Design Construction and Realization 8
3.2 Deformation Analysis 10
3.3 Boundary Condition Assumption 11
3.4 Design Model Decision 12
3.4.1 Volume Capacity 12
3.4.2 Cost Analysis of Boat Body Designs 12
3.4.3 Decision Matrix 12
3.5 Concept Evaluation For The 2nd Phase Prototype 13
3.6 Updated Embodiment 13
3.7 Updated Degisn Detail 16
3.7.1 Trash Detection 18
3.7.1.1 Backgrounds 18
3.7.1.2 Hardware used in this program 18
3.7.1.3 Equipment Set Up 18
3.7.1.4 Test Result 19
3.7.1 Other Updated Parts in Spring 20
3.8 Standards Applied 21

IV. Context and Impact 21
4.1 Economic Analysis 21
4.1.1 Budget Analysis As of The Winter Term 23
4.1.2 Budget Analysis As of The Spring Term 23
4.2 Environment Impact Analysis 24
4.3 Social Impact Analysis 24
4.4 Ethical Impact Analysis 25

V. Project Management 25
5.1 Team organization 25
5.2 Schedule and Milestones 26

VI. Discussion 26

VII. Summary and Conclusion 29

VIII. Future Work 31
8.1 Main body completion 31
8.2 Power system installation 31
8.3 Future work for spring term 31

8.3.1 Structure Reinforcement 31
8.3.2 Solar Panel and GPS auto-pilot integration 31
8.3.3 Suitable Control Method 32
8.4 Possible Future Work For Future Terms 32
8.4.1 Solar Panel Integration 32
8.4.2 Aesthetics Fine-Tuning 32
8.4.3 Additional functions 32

IX. Appendix 34
Appendix A.1: Needs and Specifications of the Stakeholder 34
Appendix A.2: Updated Needs and Specifications of the Stakeholder 34
Appendix B.1: Proposed Project Schedule For Fall Term 35
Appendix B.2: Project Schedule For The Winter Term 36
Appendix B.3: Project Schedule For The Spring Term 36
Appendix C.1: Budget List and Cost Breakdown As of The Winter Term 37
Appendix C.2: Budget List and Cost Breakdown As of The Spring Term 38
Appendix D.1: Transmitter coding (Sprint Term) 39
Appendix D.2: Receiver coding (Sprint Term) 42

X. Reference 44

I. Introduction

1.1 Background

Due to HuaYuan Construction's responsibility to pick up the trash in the Yangtze river, our group, under the guidance of Dr. Najafi and Dr. Chang, are committed to the development of an automated river cleaner All over the world, water pollution through littering can be considered a common, but high priority, problem. One of the places being affected by this strain of pollution is the Yangtze River area. As seen in the figure below, there are many cities and municipalities that line the Yangtze river.



Figure 1: Map of the Yangtze River Area

As plastic trash breaks down in rivers, the plastic degrades into tiny particles of plastic called microplastics (MP). The most likely place that these particles start would be the Three Gorges Dam that can also be seen in the previously mentioned figure. The MP float down the river and enters the East China Sea where they are eaten by unsuspecting fish and, eventually, back to us. By creating an automated river cleaner, the trash will be cleaned in both a cost-effective and efficient manner and mitigating the cause of MP in the area at its source.

This can be a great investment for local, municipal, and national governments due to the nature of the product. As gathering trash without harming the local wildlife requires a fair amount of manpower, it will require the proportionate amount of funds to ensure that the contracted company does a good job cleaning. This amount of funds would only drain the government in question's resources since this would be a recurring problem. The river cleaner being proposed will reduce the amount of money that the government facility would have to spend on cleaning the river. The price would be further reduced due to the river cleaner's lack of human interaction.

There are solutions to this situation, but there are problems with these river cleaners that can't be overlooked easily. For the river cleaner that was created by the startup known as "Urban Rivers", while this may be a robot that is, essentially, a game, the robot doesn't seem to have a large carrying capacity for trash. In addition to this con, since the program the robot runs on is power by a game server, the server, like any game, would eventually lose popularity and go offline. Not only will this produce inconsistent results for a problem that might last a substantial amount of time, but the robot would go offline almost permanently due to the game's life cycle ending. While the river cleaner, currently, in the Baltimore river known as "Mr. Trash Wheel" solves a fair amount of the problems, as well as be completely autonomous,

the trash could easily pile up at the containment booms and, from what can be interpreted from its design, the trash has to come into contact with the rotating forks in a certain way so that it can be picked up. This defect in the trash retrieval system will cause trash to pile up at the river cleaner instead of removing the trash in an efficient manner. The river cleaner being created by our group would not only have a large capacity, but would also be autonomous, be able to differentiate living animals from trash, and require little to no human interaction.

1.1.1 Updated Background

Our designed trash collection robot is considered to be working on the river that the stakeholder requested. The river has a depth of 2 meters and a width of 4 meters. The average flow rate of the river is 1.5 meters per second. The river is usually clean and neat, but the trash will appear randomly. Thus, the designed robot must be flexible enough to handle the randomly present garbage at any point during a long period of time. The garbage is usually bottles and plastic bags. Therefore, the trash collection system does not need to be very powerful but it does need to be reliable enough to collect all light garbage.

1.2 The Stakeholder's Needs and Specifications

The design of the river cleaner is heavily influenced by the needs and specifications of our stakeholder. These needs and specifications were gathered from our communication with our stakeholder and then interpreted in a way that we could arrange them into ranks of importance. Each of the needs/specifications, as well as their rankings, can be seen in Appendix A.1. Based on the rankings in Appendix A.1, the most important needs/specifications are:

- The robot is able to detect ships, living animals, and avoid collisions.
- The robot is reliable and requires minimal daily maintenance.
- Robot can charged by solar and wind system(Customer does not want this function)
- Robot can use GPS as a guide to patrol in the assigned working area (Cost way too much for the allowed budget)
- Robot is easy to control and does not require more than one person to operate

1.2.1 Updated Stakeholder's Needs and Specifications

In the ongoing communication with our stakeholder, there are a few changes which our stakeholder discussed about. In order to make the robot much easier to maintain, the solar and wind systems are removed. It takes too much time for their employers to maintain this kind of equipment with their limited education. Second, the GPS guide costs much more than what our budget allows for. In addition to this argument against the GPS system, the robot is only along the desired river, which is within a small scale of area. After reviewing this information with our stakeholder, the GPS might not be necessary and will be, therefore, removed. After communicating fully with our stakeholder, the needs and specifications are updated accordingly to their more detailed requirements. The updated specifications can be seen and ranked in Appendix A.2.

1.3 Problem Statement

As the trash in the Yangtze river reaches critical levels, the local government has decided to put a sanction to reduce the amount of trash in the river. After negotiating, HuaYuan Construction, LLC. has been tasked with clearing the trash. Seeing as this will be a very expensive venture that will cyclically recur, the client is looking for a cost efficient method that would be able to complete the objective while being environmentally conscious.

II. Methods

The design duration is three terms, and our group divided the design process into three stages. For this fall term, we mainly focused on designing the boat which will be used to be the platform of the robot and build up the model for more tests. For the winter term, the devices and energy system will be set up, and the robot's function and model product will be built in the spring term.

For this term's analysis, Cero 4.0 and Ansys 18.2 are used to build the shapes of the design ideas and to perform the deformation analysis. Our group works in the machine shop of Drexel University to build the model of the boat.

In the future work, python and matlab will be used to program and set up the robot's function.

III. Design Description

3.1 Design Idea Assumption

After careful consideration and doing research, our group comes with three design ideas for the boat body. All three models have two 0.6m × 0.2m bases and height of 0.3m. The capacity between the two bodies will be 0.275m, which is also the length of the pipe. The 3D model of three design ideas are shown in Figure 2-4 below:

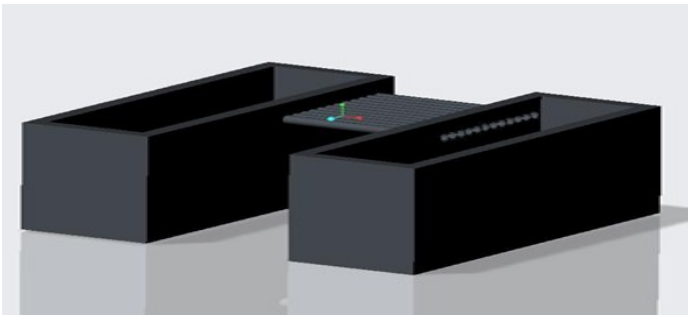


Figure 2: Design Idea 1



Figure 3: Design Idea 2



Figure 4: Design Idea 3

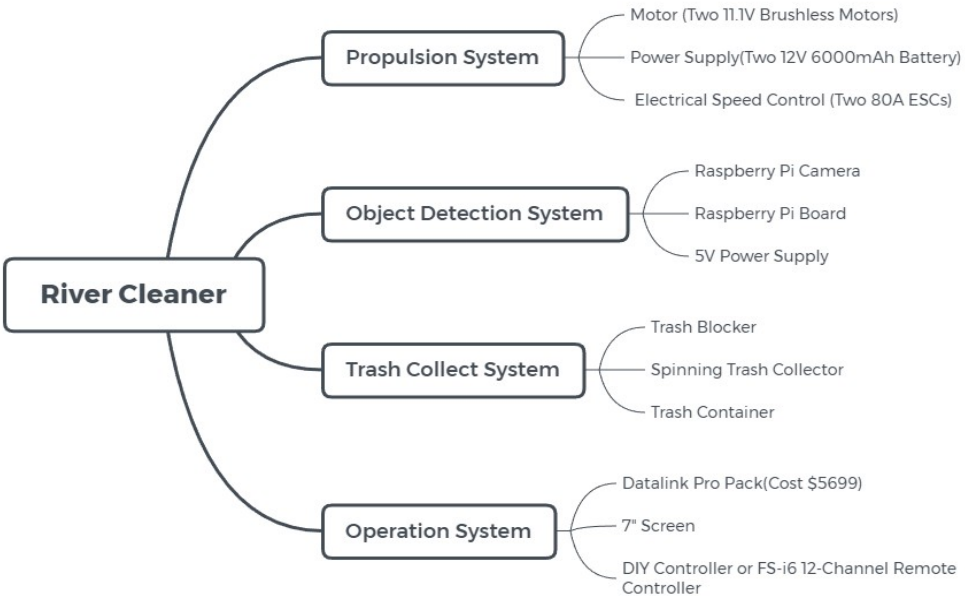


Figure 5: River Cleaner Design Diagram

The project is mainly divided into four systems as shown in the Figure 5. It contains the propulsion system, object detection system, trash collect system, and operation system. The desired working procedure has been changed. Our original design is asking the robot to observe the trash and go to collect the trash by itself. However,as it needs lots of sensors and a really good camera which can track every movement of the trash, the cost is over than our allowed budget. Sometimes there might be many pieces of trash present in the river simultaneously, so it is better to let an operator to do the controlling work.

The object detection system contains the camera, raspberry pi board, and the power supply. The power supply needs to be above 5V in case of under voltage risk.

There are the trash blocker, the rotary trash collector, and trash container in the trash collect system. One thing that needs to be mentioned is that the trash container must be easy enough to take out of the platform. In this term, several sliding rails were installed, making the trash container function like a drawer.

The overall mechanism is the raspberry camera that will detect the trash. The stream video will be relayed through the Datalink Pro Pack(or other substitute) to the mini monitor which will be taken by the operator. The operator will know the location of the garbage and control the robot to collect the trash. This mechanism costs less and it is more effective when there are

multiple pieces of trash appearing simultaneously. The stakeholder is also more satisfied with this method.

3.1.1 Design Construction and Realization

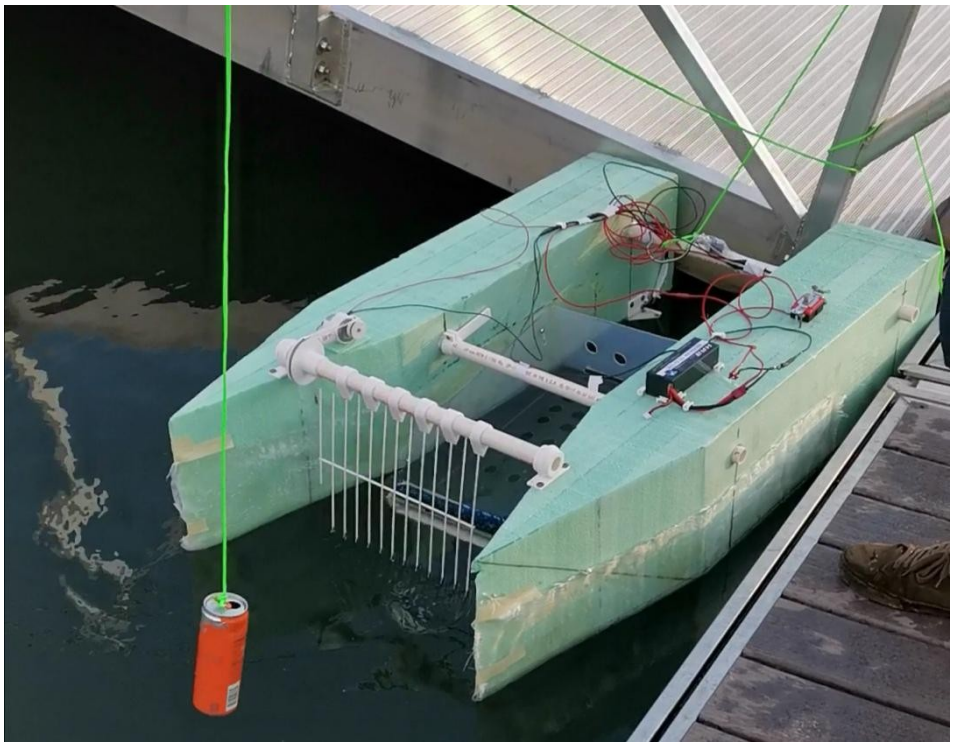


Figure 6: Designed Prototype Demonstration

As it is shown in the figure above, the prototype contains 3 sections: the floating body, garbage collection system and propulsion system. The floating body mainly has two boats which are connected by two PVC pipes. With glue inside the structure, the boat is reliable. The trash collection system locates between two boat. The core of the system are the high torque motor and the transmission system. By transporting the torque produced by the motor to the front bar, the system can rotate in controllable speed. At the same time, the steel sticks will push the garbage through the trash receptacle gate and let the trash stay in the container. The propulsion system is located in the boat's tails. It contains two powerful motors which provide boat with 24 N thrust. Even with half of the power input controlled by the remote, it is capable of not only resisting the flow of the Schuylkill river, but moving in the reverse flow at high speeds.

In the spring term, our group focused on improving the stability of the equipment. There is a brand new robot being made and the circuits are much more organized than the previous layout.

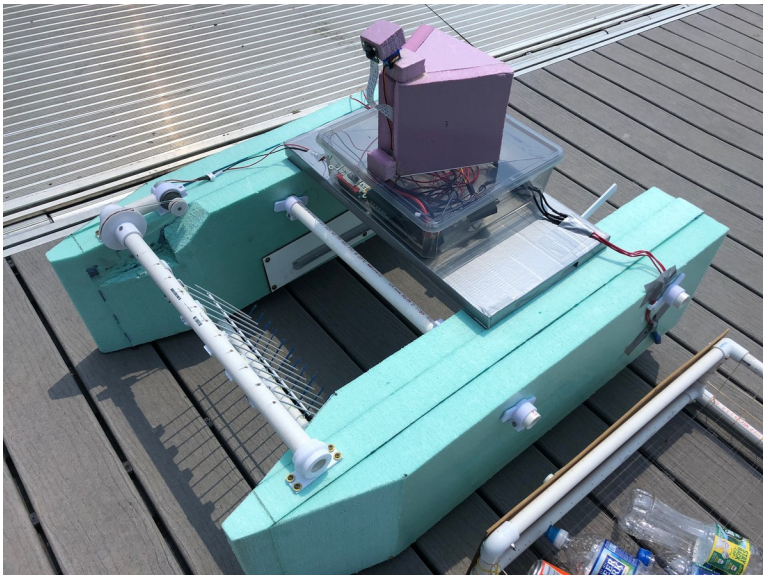


Figure 6.1 Rebuilt Robot Platform

According to customers' request, the platform needs to be more flexible. Thus, our group optimized the overall size of the platform into ⅔ of the original platform. The platform can move more freely.



Figure 6.2 Rebuilded Trash Container

As shown in Figure 6.2 above, the rebuilt trash container has a frame of PVC pipe and two sliding rails installed on each sides to make the trash container easier slide in and off the main boat for dumping the trash. A net made of fishing line can keep the collected trash inside the container. The container has a dimension of 6.5in x 25in x 16in and gave a maximum capacity of 0.045which satisfied our stakeholder's need.

3.2 Deformation Analysis

One thing that needs to be mentioned is that the model analysis that we made in the Ansys is only helping us to find out the best design of the boat platform. The boundary conditions are assumed and might be changed in the future depending on the data we get in the future.
The deformation analysis of the three design ideas are showing in Figure 7-9 below:

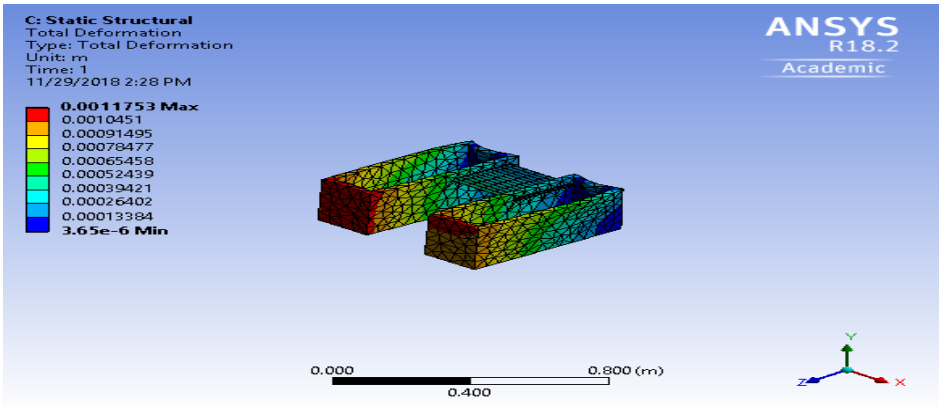


Figure 7: Deformation Analysis of Design Idea 1

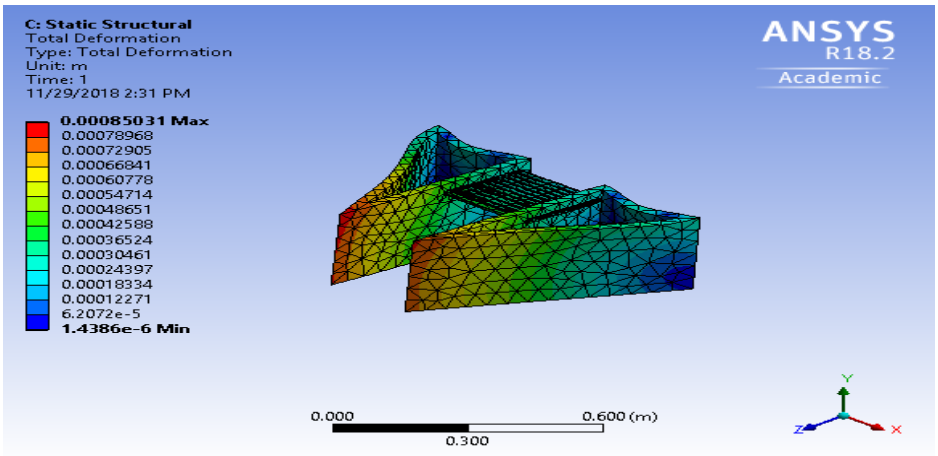


Figure 8: Deformation Analysis of Design Idea 2

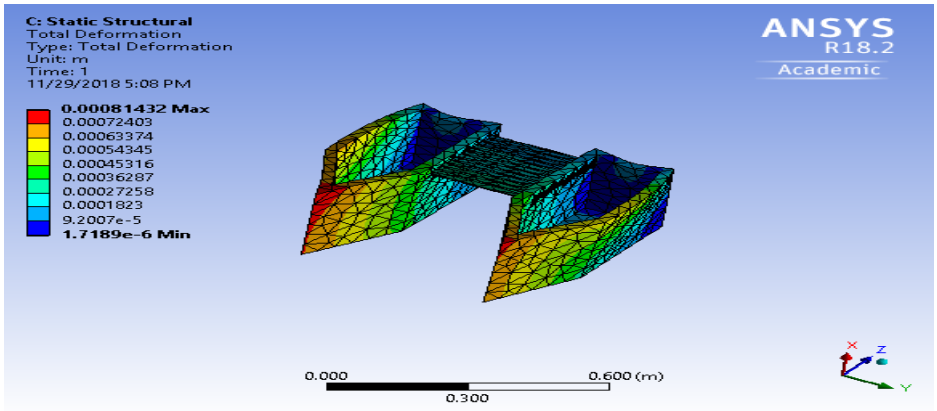


Figure 9: Deformation Analysis of Design Idea 3

3.3 Boundary Condition Assumption

The deformation analysis is based on the assumed boundary conditions.

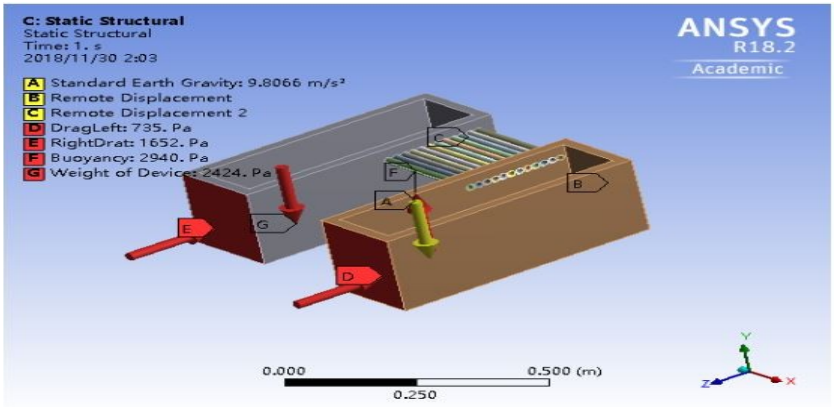


Figure 10: Boundary Conditions Application

As shown in figure 10, there is a standard earth gravity applying on the boat. There are drag forces being put in the very front surface which simulates the drag created by the water and wind resistance on the surface. Buoyancies are applied on each bottom surfaces. Also, the weight of the devices is included.

In our simulation, we assumed that the back two surfaces do not move, and by applying the drags and other forces on the other surfaces of the boat, we can see how it will deform. Figure 11 below shows the equations and data that used to calculate the parameters:

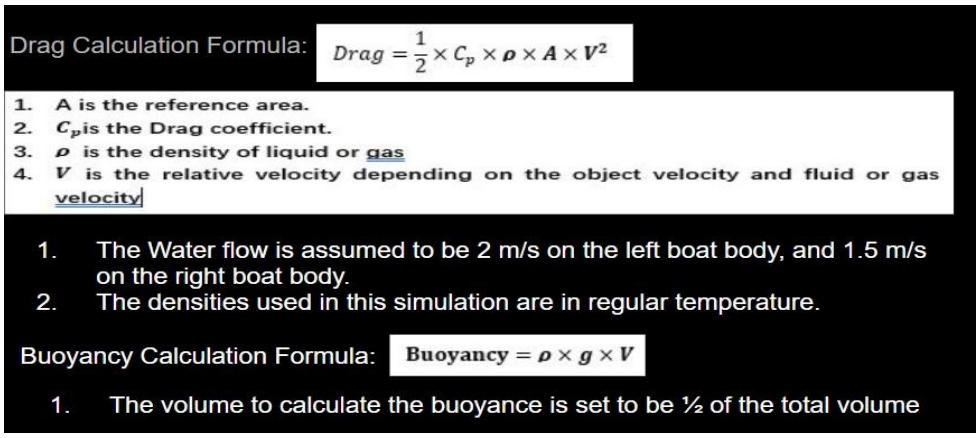


Figure 11: Drag and Buoyancy Calculation

The total weight of the devices is assumed to be 20 kg, and the temperature that is involved in this simulation is room temperature.

3.3.2 Winter Term Prototype Detailed Calculation

The total weight of the prototype now is 8 kg. The propulsion system that is used now can provide a 24 N thrust.The water flow is 1.5 m/s. The measured waterline of the prototype is 2 cm. Assume that the water flow is isotropic and same along the width of the river, whether the thrust is enough can be calculated through the first equation in figure 10 with the Cp value of 1.33. Since the boat will have a total width of 35 to 40 cm, the waterline is 2 cm; thus the Cp value that is used in this calculation is 1.33. The area that will be influenced by drag will be 0.008 and the relative velocity is 2 m/s (Maximum Stakeholders' Requirement). From the calculation, the total drag caused by the water will be 21.28 N while the two motors can provide a thrust of 24 N. This shows the propulsion system works well and the propulsion meets the stakeholder's requirements. In work that will be done in the

Spring term, two more motors will be installed to boost the thrust and give the boat a much higher propulsion. Also, during the test in the Winter term, the prototype is cruising at a velocity of 1 m/s for safety consideration as the control of the river cleaner's movements become unstable.

3.4 Design Model Decision

3.4.1 Volume Capacity

Table 1 below shows the volume capacity of the boat main body.

Table 1: Volume Capacity of Three Design Ideas

	Design 1	Design 2	Design 3
Volume Capacity Equation	V=w×l×h	V=Bottom Area×height	V=Bottom Area×height
Volume Capacity(one boat)	0.02475	0.008859	0.01863

3.4.2 Cost Analysis of Boat Body Designs

The cost analysis of three design ideas is shown in Table 2 below:

Table 2: Cost Analysis of Three Design Ideas

Form cost		Pipe cost			Design 1	Design 2	Design 3
thickness (in)	1	size (in)	0.5	volume of boat base (m^3)	0.027216	0.023556	0.022664
wide (in)	24	length (in)	120	form cost ($)	17.24	14.92	14.36
length (in)	24	length (m)	3.048	length of pipe (m)	3.30	3.30	3.30
volume (in^3)	576	cost/ pipe($)	2.31	pipe cost ($)	0.03	0.03	0.03
volume (m^3)	0.009439	cost/ meter($/m)	0.76	total cost ($)	17.27	14.95	14.39
cost per form($)	5.98						
cost per unit($/m^3)	633.55						

3.4.3 Decision Matrix

Table 3 below is the decision matrix:

Table 3: Boat Body Design Decision Matrix

Boat Body Decision Matrix							
Needs	Weight	Disign 1	Disign 2	Disign 3	Score 1	Score 2	Score 3
Boat body needs to be stable	9	8	7	9	72	63	81
Boat needs to have a big capacity	8	10	6	9	80	48	72
Boat's cost should be reasonable	7	3	5	7	21	35	49
Total Score					173	146	202

From all the data and analysis provided, our group is going to choose a best one for the design and start to build the model.

3.5 Concept Evaluation For The 2nd Phase Prototype

First, the prototype made in the Winter term shows the target users a better understanding of the product. In the Fall term's description, many mistakes were made that the users may not be able to easily understand the mechanism of product. However, this term, the major functions are shown on the prototype (although many parts are rough).

The change in the design is shown on the prototype as well. During the Fall term, the group planned to use a transport belt to collect the garbage. For the Winter term, we used a garbage sweeper system after discussion and evaluation. The main frame is changed too as shown in figure.

Generally, the prototype meets the expectation. Most of the mechanical parts work properly. The garbage sweeper took the group significant effort to function correctly. The transmission belt between motor and garbage sweeper pulley did not cooperate well in the beginning. However, that was resolved by adding sandpaper to the pulley wheels to add more friction. There are some defects that did not meet expectations too. One is the exposed wires. Most of the wires are exposed outside the main body, making it vulnerable to damage from the environment and visually unappealing. The clearance between parts were not considered fully before assembly, makes some of the parts conflict with each other.

3.6 Updated Embodiment

The skeletal frame is made of PVC pipes while the main body is made of styrofoam glued together with epoxy resin. Connection of pvc pipes are comprised of 3D printed parts as shown in the figure below, bolts, and stainless steel rods.



Figure 12: 3D Printed Connector Pieces

The two motors (4 motors for final product) are placed under the floating body with 3D printed mounts. The electronic speed controllers are placed by the wall of floating bodies temporarily with tape (a formal design will be used for the final product). The garbage sweeper is made of steel rods and a PVC shaft connected by 3D printed connectors. The camera mount is fully 3D printed. The group used a smaller battery pack than the the one used for the actual product to ensure the safety of experiments. The prototype is controlled by a remote controller instead of microcomputer, since the group decided to do the programming in the Spring term.

The material utilized on the final product is generally the same as the prototype. Instead of using a plastic container as the trash tank, we used PVC pipes to make the frame, and fishing lines to make a web to hold the trash. On left and right sides of the trash tank, we installed the rails from desk drawers, so the tank can be drew out easily for replacement as shown in the figure below.



Figure 13: Trash tank with rails on sides

To protect the propeller when putting the product on the ground or in a vehicle, stainless steel wire is used to make a stand. So whatever the position on the ground, the propeller would be lifted 1.5in from the ground.



Figure 14: Propeller with stainless steel wire protection

After discussing, an arduino mega controller is used for the core of controlling system. Other controller boards are utilized for different functions too. Instead of using a 6-channel remote controller, a programmable DIY controller is used.

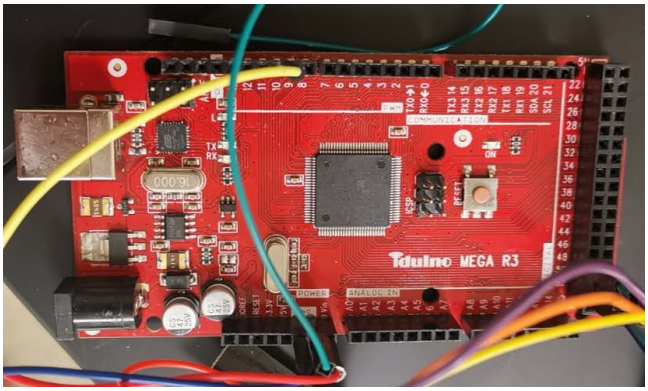


Figure 15: Arduino MEGA

The arduino MEGA is mainly used to control the motor' speed. It receives PWM signal and transport the operater's command.



Figure 16: DIY Remote Controller

This is the DIY controller that is going to be used in the test. The joystick can control two motors simultaneously. Also, the speed control can be achieved by the spinning controller. Through programming, the 'going left' command can be translated into increasing the speed of the right side motor, and at the same time, lower the speed of the left side motor. One thing that needs to be mentioned is that since brushless motors are used in this project, it is impossible to give robot reverse thrust command. In the future, this function can be completed by using DC motors.

3.7 Updated Degisn Detail

The prototype can be divided into 4 sections: frame, floating body, garbage collection system, and propelling system.

Frame is made of PVC pipes, as shown in the previous section. In the beginning, the group planned to use aluminum pipes for high strength. After evaluation, PVC pipes are adopted because it already has enough strength, while the weight is lighter. The connection parts are 3D printed with many versions of improvement. the previous versions are proved unpractical or fragile. The connection between the main body and the sensor mount is under a "quick release" design, in order to be convenient for users to put in vehicles (fig 12).

The floating body is made of glued styrofoam. Before choosing this, options such as aluminum and wooden shells were discussed, but abandoned for multiple reasons. The group decided to reinforce the surface of the floating body with fiberglass cloth and epoxy resin in the beginning, but found the idea not practical during the process.

The design of the garbage collection system changed during the Winter term. It was, originally, a transmission belt system that would be used to pick up the trash. This method was later exchanged with a sweeper pulley system, as shown in previous section, for its simplicity and decreased room for errors. There is a small-scale transmission belt between the motor and the sweeper pulley. These parts would be held in place by the part shown in the figure below. The friction between belt and pulley was a significant problem. In order to increase the friction, sand paper was cutted into strips and attached to the pulley wheels. This ensured that the friction was enough to transmit the torque from the motor to the pulley system.



Figure 17: 3D Printed Pulley Mount System

The main components of propulsion system are the brushless motors. The group utilized 2 motors instead of 4 on the prototype in order to achieve a calm and ideal experiment environment. The wire soldering is a detail that's worth attention as it took a longer time than estimated. Concealing the wires is mostly done with duct tape and silicone glue; this issue will be addressed in the final product. The circuit diagram used is represented in the figure below.

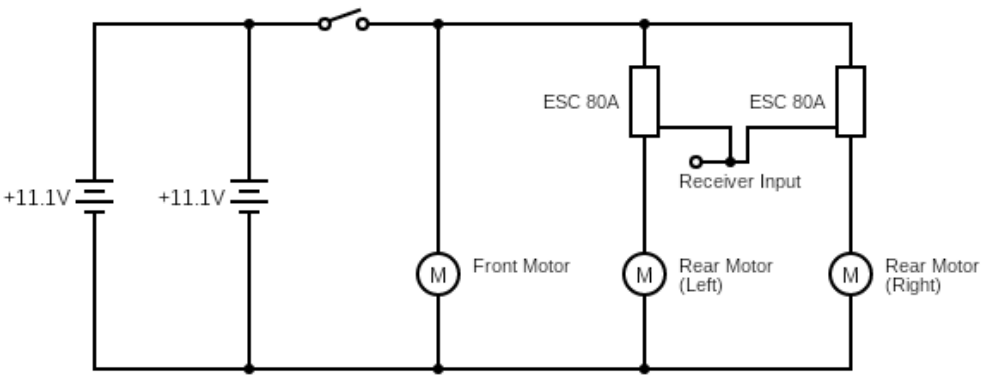


Figure 18: Current Circuit Diagram For Powering Up Motors

Two 11.1V 6000mAh 50C Lipo batteries are in parallel as the power supply. A toggle switch is used to turn the power on and off. All motors are in parallel for gaining the same 11.1V from power supply. The front motor is constantly running while the two rear motors are connected to two ESC's (Electronic Speed Control). The ESC reads signals from a receiver to achieve manual remote control. When the GPS auto-pilot system is integrated into the robot, the circuit diagram will be subjected to change.

3.7.1 Trash Detection

In this term, two of our group members worked on the trash detection system. This function takes a lot of time to set up the model identification. Fortunately, the object detection system is finished. In this part, the background, equipment set up, and test results will be explained.

3.7.1.1 Backgrounds

In this program, TensorFlow is mainly used to detect and identity object. TensorFlow is an open-source software library which is used to analyze data flow. It can be used in machine learning as well. In the river cleaner, the robot needs to recognize the object and identify it as trash, alive animal, or obstacle. Once the object is recognized by the robot, the robot will start to do different actions. For example, if one object is defined as bottle, the operator will see the notification in the operating screen, and he/she will control the robot and go for the bottle.

3.7.1.2 Hardware used in this program

There are several hardwares that being used to achieve this function. It is listed in the following table.

Table 4: Materials Bought And Corresponding Price

Item	Number	Single Price
Rasberry Pi	1	$ 37
Rasberry Pi Camera	1	$ 32
HDMI Wire		None
Monitor	1	None
Power Supply(5 V)	1	$ 7.99

3.7.1.3 Equipment Set Up

As the equipment setup is unnecessarily complicated, the overall set up will be introduced in this report.

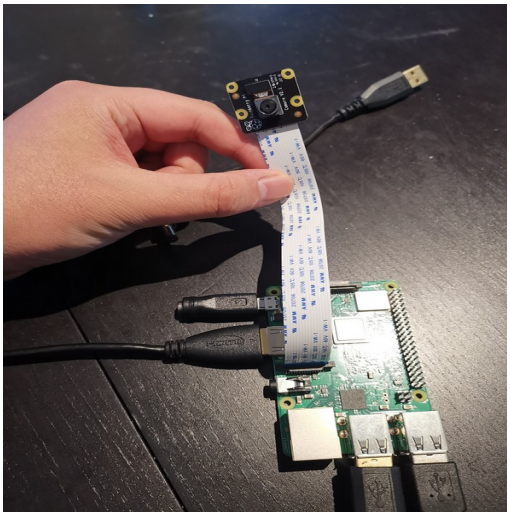


Figure 19: Raspberry Pi Camera Configuration

As you can see from the figure, the Raspberry Pi camera is connected to the Raspberry Pi board, and the upper USB can be connected to the 5V power supply. In the testing phase, one HDMI wire is used to transfer the video from camera to monitor, but in the real field, the video will be transferred to the operator screen.

3.7.1.4 Test Result

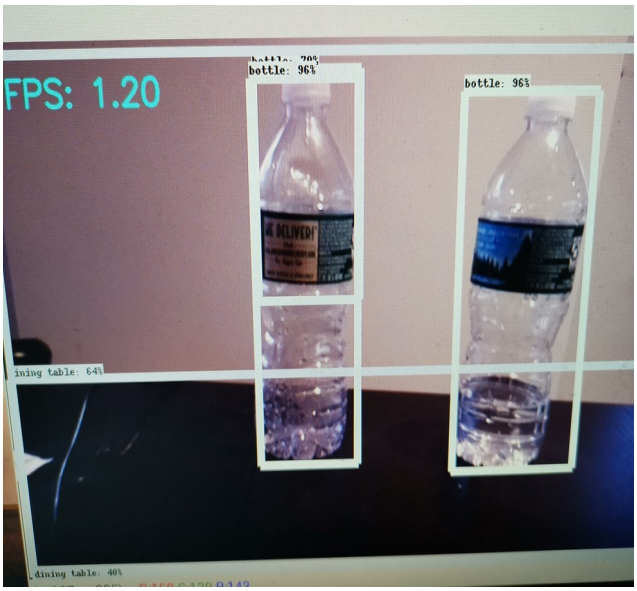


Figure 20: Object Identification Test Result

As you can see in the figure, two bottles are recognized by the Raspberry Pi. There are identification boxes and percentage confidence on the upper corner of said boxes. In this figure, the confidence of the recognition is above 90 percent, which means it is a very accurate detection. In order to test the reliability of the object detection, more tests are being done and our group will collect all the data which will be used for later analysis. The problem of the current detection system is the low FPS. As you can see in the figure, the FPS is 1.2. During the test, the FPS is usually around 1 to 1.5. Since the robot is working in a river with a flow rate of 1 to 2 m/s, the camera might not be enough to detect the moving bottles. Thus, the camera needs to be updated within a reasonable range of cost.

3.7.1 Other Updated Parts in Spring

After the tests done last quarter, a series of problems were exposed. In order to fix those problems, improvements are applied on the final product.

As shown in figure 17, the trash sweeper motor mount is fixed to the main body with nails. It was left free without limitation last quarter. This improvement would prevent the tension force of the transmission belt from pulling the motor away from its position.



Figure 21: Rails on the floating body

Rails are fixed on both the trash tank and main body (figures 13 & 21). The rails are inclined downwards, so the trash tank could stay at its position under gravity. The difficulty of installing this part is that the distance between two floating body must be precisely measured.



Figure 22: Inclined Trash Trapping brush

The trash trapping brush is improved too. Instead of using a whole piece of the brush, small bundles of plastic hairs are fixed on a PVC pipe (it traps trash inside the tank) as seen in figure 22. The bundles are directed towards the back of the boat to let the trash enter smoothly. The bundles are located between the rods of sweeper, so less friction between the sweeper and the brush exist. After assembly, the product will be ready for testing (figure 23).



Figure 23: Assembled Product (without electronic parts)

3.8 Standards Applied

As the river cleaner is going to be used by companies and cities alike, the robot's construction has to follow standards based on its components. Since it has electrical components, it would follow the IEEE standards to ensure no one gets shocked or that the robot short circuits. In addition to the IEEE standards, we also use the ASME standards in the conceptualization of the robot as a whole and the individual components. The ASME standards were also implemented in the construction process of the robot and component analysis.

IV. Context and Impact

4.1 Economic Analysis

One of the targets of this project is to decrease the cost of river cleaning, so economic analysis is necessary.

Table 5: Predicted Cost List of the prototype

Part name	Count	Cost each
Floating material	1	15
Structure tube	1	8
Arduino control system	1	40
Motor	2	6
Propeller	2	2
Motor controller	2	4
2.4GHz remote controller	1	30
Battery pack	1	20

Camera and processor	1	50
LIDA system	1	100
Total		287

Table 6: Cost List of the Final Product

Part name	Count	Cost each	Note
Floating body (customized)	2	100	Made of fiberglass and metal frame
Structure Material	1	20	Improved with 3D printing
Arduino control system	1	0	Come from prototype
Motor	2	20	
Propeller	2	4	
Motor controller	2	15	
2.4GHz remote controller	1	0	Come from prototype
Battery pack	1	100	
Camera and processor	1	0	Come from prototype
LIDA system	1	0	Come from prototype
Solar panel	1	60	
Solar panel managing system	1	30	
Tools	1	200	Estimation
Total		688	

In order to avoid high costs, we chose existing and mature parts on the market (like the LIDA and graph processing system). The cost of final product would be controlled at an acceptable level.

The following table is the details of river cleaner:

Table 7: Details of river cleaner product

Product Estimate Price	Lifespan	Maintenance	Interest Rate	Salvage Value
$1000	5 years	$250	4%	$150

The equation used to calculate the annual cash flow is as follows:

From the equation above and the data provided, the annual cash flow is $447, and the cumulative cash flow $2235 for it's entire lifespan.

The hourly rate of workers is $4/hour in China and $12/hour in USA. Assume that they work 4 hours per day and 100 days per year. The following is the cost of both methods:

Table 8:Cost of Methods of Collecting Garbages

	Annual Cash Flow	Five Year Total Cost

Work Done by River Cleaner	$447	$2235
Work Done by Hiring Workers in China	$1600	$8000
Work Done by Hiring Workers in USA	$4800	$24000

4.1.1 Budget Analysis As of The Winter Term

In order to make the initial prototype, many new parts are purchased through internet or local sellers. The main parts of this prototype is motors and batteries. For the motor, members of our group brought four brushless motors from China over winter break. For the battery, two HRB 6000 mAh, 11.1V batteries are chosen. Along with the batteries, one charger was purchased as well. After everyone met up again when the term started, more materials were purchased throughout the term. To prevent getting damages from the collision, fiberglass cloth is used to cover on the boat. In order to make the fiberglass cloth stick on the boat, the following coating materials are used. This would allow the fiberglass cloth to gain a hardened shell and have waterproofing.

For this term, the prototype is mainly operated through a remote controller to control the motion of the boat. The motors that were bought from China were used for the boat's propulsion in a pairs to make it easier to control. For the garbage collection system, a high torque gear box electric motor is used to rotate the trash collect "fence" and one solid round belt is used to transport the torque from the motor to the trash collect bar. Since the 3D printed parts didn't have enough friction to have the torque transferred from the gearbox to the front bar, small slivers of sandpaper from the machine shop were added to provide that necessary traction. The budget list for all materials bought up to the Winter term, as well as the remaining budget after the total costs have been considered, are presented in Appendix C.1. As shown in this appendix, we still have a almost half of the specified budget left to make any modifications and additions to the river cleaner. In this respect, the goal of meeting the budget is going well for now.

4.1.2 Budget Analysis As of The Spring Term

As the programming for the trash detection system and the new remote required more components, the required materials were purchased through Amazon and arrived in a short amount of time. The factor that assisted in saving a significant amount of funds was that there were still materials leftover from the previous term to rebuild the mechanical portion of the river cleaner.

For this term, fixing the conflicts that were discovered from the test done in the previous term took priority over the secondary objectives as these problems needed to be resolved for the robot to run successfully under the stakeholder's specifications. The new material price and how they affected the remaining budget are shown in Appendix C.2. As seen in said appendix, the new purchases have reduced the remaining funds by half. This shows that the final product remains significantly within the allowed budget.

4.2 Environment Impact Analysis

The project is designed to clean the surface of the river, so the overall impact to the environment is positive.

The first part is positive impacts. By applying this product, the surface of the river in a high-population dense area would improve significantly. River cleaning workers mostly use gasoline or diesel powered boats, while our project uses electricity generated by solar and wind energy. Compared to the existing solution, the product would significantly decrease the amount of pollution caused from completing the task. Since this product does not need to carry a driver, the efficiency of boat would be significantly higher.

The other part is negative impacts. Although the overall impact is positive, side effects exist as well. The battery pack is a source of pollution. Lithium battery recycling is becoming a mature technology nowadays, but a small amount of pollution still exists. The critical material of solar panels is silicon, the production of solar panels would produce small amounts of toxic liquid and gas. These side effects are already under strict control by governments. The product does not generate pollution during its lifespan, but all of these side effects happen before or after the service period of the product.

4.3 Social Impact Analysis

Since the product uses mature and abundant parts from the market, the production would not change the supply-demand relationship of materials. Since the product can not cruise in a large area of water, the overall demand of the product would not be high enough to influence the supply market.

The application of this product may cause a decrease in demand for river cleaning workers. However, this would not cause serious unemployment. Since the typical river cleaner employment period is a seasonal position, it makes using that line of work as a steady source of income difficult. The amount of river cleaners is low nowadays, which can not effectively eliminate the garbage on the water's surface.

After applying this product, the aqueous environment in high-population dense districts would improve. Any garbage on the water would cause the "Broken Window Effect", making people be less careful about harming the environment through littering the rivers.

4.4 Ethical Impact Analysis

The product helps people to improve the environment, creating better living circumstances. At the same time, it will arise the sense of environment protection within the society. As long as the river cleaner keeps collecting the trashes in the river, the lake, and the sea. More and more animals can have a better environment.

V. Project Management

Dr. B.C.Chang and Dr. Najafi will give us instructions about our designing system and our group has regular meetings with them. Since our group has mechanical engineering students and an electrical engineering student, we can work on different aspects of the project and communicate with each other. All the designing drafts, references, and other files will be

shared on onedrive for group members to access. Groupme will be utilized for communication between members. Group members will contact stakeholders either by email, phone or communication apps. All the construction of the system will be proceeded in drexel's lab and machine shop. All the cost of the project will be recorded and shared in the onedrive. To ensure that we stay on track to complete the product on time, we agreed upon a project timeline that would allow us time if something were to go wrong. This schedule can be viewed in Appendix B.1. Since the schedule conceived and agreed upon stretches until late into the spring term, what is shown in Appendix B.1 is the schedule for the Fall term and some of the winter break to show the physical research category.

5.1 Team organization

The team role and responsibility is described in the following table:

Table 9: Role Delegations and Specific Roles

Name	Role	Specification
Tongming Huang	Mechanical Designer	Platform Design and Test
Chun Leung Devon Liu	Project Manager	Financial Record Keeper, Keeps Group on Track, and Platform Design
Xiao Tan	Mechanical Designer	Structure Design
Yu Zheng	Electrical Engineer	System Programing and Circuit Design
Jie Xu	Project Manager	Quality Control and System Improvement

From the table, it shows clearly the separation of work with few areas of overlap. Basically, the project is based on different parts. Yu Zheng is better on electrical engineering and he can do the propulsion system. Jie Xu is better on quality control and system improvement.

The tools that are used in this project are summarized in the following table:

Table 10: Resources Used For Progressing The River Cleaner

Function	Tools
Team Communication	Groupme, Wechat
File Storage	Google Drive, Onedrive
Model Design and Analysis	AutoCAD, Ansys 18.2
Online and Local Purchase	Amazon, Ebay, Home Depot, Lowe's

5.2 Schedule and Milestones

As viewed by Appendix B.2, the Winter term was mainly used for construction, testing, and troubleshooting the river cleaner. The first milestone that was reached was the river cleaner succeeding in manual trash collection and movement in the desired environment. This is considered a major milestone since this meets some of the client's specifications and shows that it is functional as it is now. While this is a huge step forward, the other specifications must be addressed. The next major milestones would most likely be the complete integration of the GPS auto-pilot system and installation/incorporation of solar panels and necessary sensors into the robot. The final major milestone will be the completion of the river with all requested specifications met. While making the robot look presentable is a milestone, it will be considered a minor milestone since the deadline for that requirement is flexible.

Represented in Appendix B.3, the Spring Term was utilized for programming, construction, testing, and troubleshooting. The first milestone, equation conceptualization, was finished early in the term as it was started in the winter term. As of the spring term, tuning the physical design and integrating the object identification system were considered as one milestone as they coincide with each other. This milestone was completed quite recently as the object identification system integration and structure tuning were completed with ease. The aesthetics still stand as a future work milestone, but is considered a tertiary objective since the design, with the other future works, is subject to flexible due dates.

VI. Discussion

Our group did 100 tests on bottle recognition, and the following the data scatter graph.

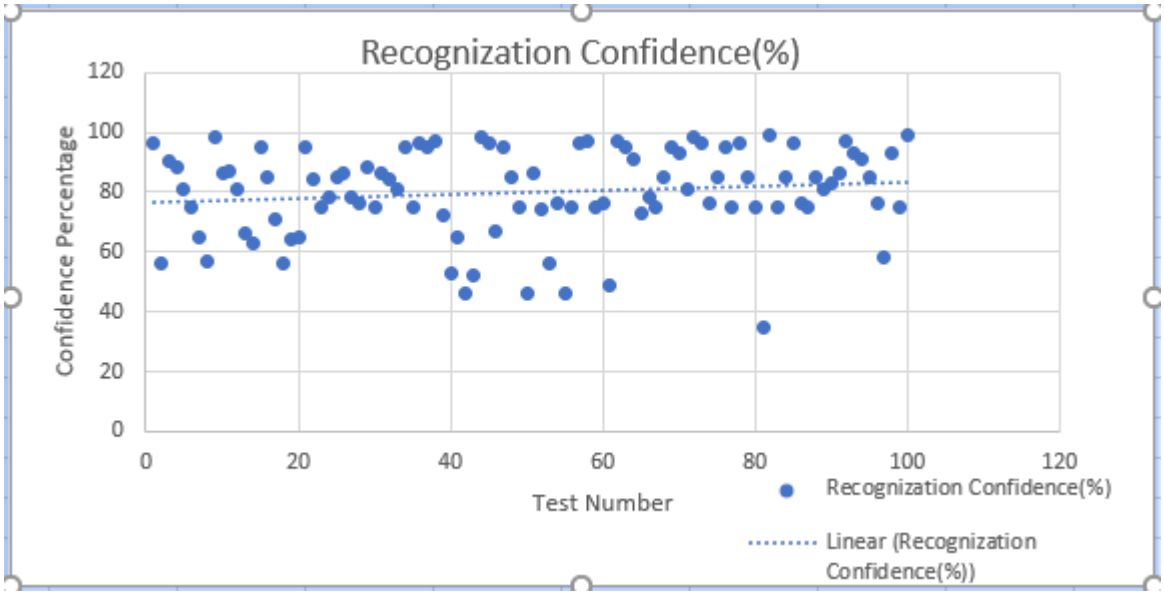


Figure 24: Bottle Recognization Confidence Test Result

As shown in the figure above, the confidence of bottle recognition is decently consistent. The average value is around 80 percent, which gives a good recognition feedback by using our trained model. There are some tests where the confidence is lower than 60 percent. The possible reason might be the tester moved the bottle too fast which goes above the camera's frame rate. The camera needs to be updated in the future.

As it is shown in the economics analysis, the cost of fives years usage of the river cleaner robot is $2,235. At the same time, it is $24,000 hiring workers to do the same job in the United States. Moreover, the robot can work in some extreme environments, while workers' safety needs to be considered. As long as regular maintenance is ensured, and the robot can work on schedule, it can relieve society of a huge funds sink. Additionally, by collecting the garbage, the environment becomes better for humans and other animals to live in.

From our deformation analysis, the deformations for all three designs are very small. The reason we choose design three is that it has the best deform performance and, at the same time, it has the lowest cost. However, it has less volume capacity and can not hold as many devices as design 1. In the future, if more volume capacity is required, we might consider to change the design since the deformation of all the design ideas are small, and will not be much different in reality. For the cost, if we need to produce more models in the next several terms, we can discuss with the providers about the price of the materials and get a better cost of the materials.

During the test in the Winter term, the river cleaner successfully collected water bottles from the river as per one of the desired tests. However, several problems with the sweeper mechanism were found during the test. Firstly The conveyor belt kept slipping from the wheel of the gearbox. This problem was solved by adding sandpaper to the surface of the wheel to increase the friction. Secondly, in our current prototype, we simply turned the screw to fix the trash receptacle to the floating body. While removing the tank from the floating body frequently, the screws attached to the foam became loose, so the tank can not completely connect to the floating body. Thirdly, the propulsion system is not stable as it is now. Lastly, the steel sticks of the trash collection system need to be replaced or improved due to the water bottle getting stuck between the steel sticks and the boat. This would drastically reduce trash collection efficiency and/or break the mechanism. In spite of these setbacks, most of these problem were eventually resolved and the progress of the river cleaner hit its first milestone. The circuit is exposed to rain and splashes from the boat's movement through the river. A hard cover will be needed to prevent water damaging the circuit. Two batteries are in parallel connections. This may cause a huge current between the two batteries when one of the batteries is low or empty and the other one is fully charged, causing the fully charged battery to start charging the other battery. This can be prevented by adding a diode on the circuit. The battery charger also needs to be wired in parallel so that both batteries can be charged simultaneously.

The control of the cleaner was manual by using a remote controller. The receiver for the remote controller sends a PWM signal to the ESC in order to control the speed of the motors. The PWM signal has a frequency of 50 Hz with the impulse length varying from 1ms (motors off) to 2ms (full throttle). Once the automation replaces the manual control, this control system will need to generate same PWM signal to control the motors.

The mathematical model of automation was not completed. In the process of doing state-space analysis, the angle of the machine was a problem that will continue to persist into the Spring term. The following is the basic setup of mathematical model:

The assumptions being made are the cleaner is a moving object on a 2 dimensional surface, the target trash is at the origin (0,0), and the input is u.

Constants:

k: percentage of motor force is used

F: maximum force motor provides in the river
m: mass of the cleaner machine
State Variables:
x1: x-coordinate of relative position between cleaner and the target trash
x2: velocity along the x-axis
x3: y-coordinate of relative position between cleaner and the target trash
x4: velocity along the y-axis
Output:
y1: x-coordinate of relative position between cleaner and the target trash
y2: y-coordinate of relative position between cleaner and the target trash

The general form of state-space equations for a linear system are as follows:

,
,
,
,
,

Since is not a constant, it is not possible at the moment to retrieve matrix A from above equations. This mathematical model need further improvements next term. Once this basic mathematical model is done, more conditions will be considered, such as river waves, river flow speed, and other calculation or estimation errors.

The image recognition system is working within 5 meters on the ground or table. When the trash is floating on the river, it is very difficult for it to recognize due to the light reflection and other conditions. This requires the system to be retrained with data of objects on the river. The process of collecting this data and retraining the model will be at least several months. The automation requires an input from the image recognition system so that it can pass commands to the arduino to achieve automation. Hence, this image recognition model and the automation model will be remained for future works.

The manual control system is changed to a combination of an arduino mega, a receiver, and a transmitter. The transmitter is constantly reading a joystick and a potentiometer. The potentiometer controls the output force of both rear motors. The joystick controls the output force difference of the two rear motors. When the joystick has a displacement towards left/right hand side, the river cleaner will make a left/right turn. If the joystick has a displacement downwards or no displacement, the river cleaner will be idle at its current position. This manual control is easier than the previous ready-to-use control as the user can just control the cleaner by directions instead of rotational speed of two rear motors, which satisfies stakeholders' needs. The code used for the transmitter and receiver are viewed in Appendix D.1 and Appendix D.2.

The main challenge of using this new manual control is reading the joystick. The joystick simply outputs x and y values from its x-y coordinates. Since our motors cannot rotate in reverse direction, any readings of y value below 127 (y at origin) will make the cleaner idle. For all the y value above 127, there are 3 conditions. The first condition is that the joystick is completely towards the y axis. This will make two motors running at same speed. The second condition is that x is below 127. This means the cleaner making a left turn. The third condition

is that x is above 127 which is a right turn. Both turning conditions are hard to program. The output force difference of the two motors is hard to decide. As the joystick moves on a half circle from -90° to +90° clockwise, it involves two conditions. For the first quarter of the circle, the right motor should stay at the same power but the left motor output force should increase. For the second quarter of the circle, the left motor should stay at same power but the right motor output force should increase. The output of the two motors has to deal with the angle and the distance from the origin.

VII. Summary and Conclusion

From the data and figures listed, the river cleaner is both economically and environmentally friendly. By using the river cleaning robot, citizens will live in a more pleasant environment. The government no longer needs to hire workers to deal with the garbage in rivers and it can save a lot of their budget.

According to the progress of our senior design in the Winter term, the goal of this term was reached and the progress of the river, as deemed by the project schedule, is completed on schedule. The prototype accomplished the task of collecting trash in the river. The problems that were found during the test in the winter term were solved in the spring term. The major change of the main structure is the design of a drawer tank; this helps the user to remove the tank from the boat with ease. During the test, the robot can collect trash in the river efficiently and the tank has a 0.045 capacity which allows the robot to collect bottles at a sufficient rate. According to the test, the boat has enough speed to work in the water, the speed can be controlled between 1-2 m/s with the speed control system.

Although the automation is not achieved yet, the manual control is improved for user-friendly purpose. The automation system development will certainly go beyond stakeholder's budget due to the needs of large amount of data in addition to computation resources. As this functionality stands as a secondary objective, this is an allowable omission for now. The battery can not support the robot working 4 hours for full charge, but a solar panel system will be added to the system in future work to meet stakeholders need.

Table 11: Stakeholder's Need & Design Specification

Needs	Numerical Value	Design Specification
Efficiently collects trash	<30 seconds (10 meters distance between trash and robot)	Meet
Easy to maintain	< 1 hour of maintainance time per week	Meet
Within reasonable budget	$1000	$704
High garbage capacity on one charge	.04 m^3	0.045m^3
Can cruise at a stable velocity in the	1-2 m/s	1m/s

river		
Working time for full charge	4 hours	NA
Easy movement on the river	near 0 turning radius	0

VIII. Future Work

8.1 Main body completion

The main body of the robot prototype should be completed at the beginning of the winter term. Analysis and tests will be performed to choose motors and to decide a precise location to mount the motors. The total weight of the prototype will be around 20 kilograms. More materials will be used to add support for the solar and wind power system. The main functionality of how to collect the garbage in the river may also be decided in this stage.

8.2 Power system installation

A 12V solar-wind combination power system will be designed. Calculations will be done to make sure the designed 12V power system is able to power up the motors, GPS systems, sensors, microcontrollers, and microprocessors. The solar power system will be tested first. A 12V 50-Watt Polycrystalline solar panel and one, or two, 12V 9Ah Sealed Lead Acid battery will be chosen. A 12V 9A PWM charge controller may be also chosen. The solar panel, PWM charge controller, and the motors will be properly wired and tested in an outdoor environment prior to installation. The motors will also be 12V high torque with estimated 8000 RPM. This system and motors will also be tested if it provides enough force to the robot in the river. The wind system will also be powered by a 12V wind turbine. Since the wind turbine normally has more wattage than the solar panel, the wind system may be chosen to be higher than 50-Watt.

8.3 Future work for spring term

8.3.1 Structure Reinforcement

The floating body of the boat needs to be reinforced since the foam is not strong enough to hold the trash tank between the floating body. A piece of styrofoam will be glued to each inner face of the floating body. The trash tank will also be replaced. A “drawer” trash tank will be used in the Spring term to make trash removal easier. The pulley holder of the “drawer” will be fixed to each styrofoam of the floating body. The steel sticks of the collecting system will also be replaced because during the test in the Winter term, our group found that sometimes the water bottle will get stuck between the steel sticks and the tank.

8.3.2 Solar Panel and GPS auto-pilot integration

As per the specifications shown in Appendix A.2, the river cleaner must be able to work autonomously in an assigned working area and utilize solar energy to charge the battery. During the Winter term, a differential equation for the system was conceptualized as the basis of this system. Around this equation, a program will be developed and tested using the robot as the test subject. Solar panels will be purchased and integrated in a way that it flows with the rest of the circuit.

8.3.3 Suitable Control Method

During one of the tests initiated in the Winter term, the tester found it was hard to control the boat's movements. In the Spring term, a mathematical model will be improved and put into use through a program. Matlab and python will be used to program. Fortunately, the component needed for the "brain" of this control system was already purchased. Most, if not all, of the term will be used to work on this task and the GPS auto-pilot system since these will be need to work in tandem to function properly.

8.4 Possible Future Work For Future Terms

8.4.1 Solar Panel Integration

As the construction of the autopilot system was in its programming phase. It was determined that in order for the autopilot system to be completed on time, the solar panel integration must be moved to possible future works. There were various factors that supported this decision such as the possibility that the panels would not go through enough testing before, and after, the integration process to be fully operational and the fact that this objective was originally one of the low -ranked primary objectives in the specifications requested.

8.4.2 Aesthetics Fine-Tuning

Due to the difficulty of the task at hand, the functionality of the river cleaner held significantly more importance than how it looked. While we had a significant amount of funds remaining from the previous term to do whatever was needed, the majority of the attention was directed towards the autopilot system while the remaining was towards fixing the more immediate conflicts that were derived from the test done in the previous term. In future terms, it would be desirable to make the river cleaner look more aesthetically pleasing without disrupting the robots functionality. Since this robot will be sent to rivers that would, most likely, have people near them, the robot needs to look nice if it were to be associated with our client.

8.4.3 Additional functions

After discussing about the current functions (including those in progress), we found the platform itself has sufficient room for more functions. A GPS module would allow it autonomous functionality when combined with the object detection system. Another example is water quality detection. By adding a module of water sample collector, the product can go to desired locations to pick up water samples. The operators would only need to analyze the samples after the product returns; this would save time for operators. Another example is crucial area rescue. Since the product is able to recognize objects floating on water, it would also be able to recognize people after modification. The product can cruise in an area of high drowning risk for long periods of time. Once it recognizes the help alarm, it would use its maximum power to reach the people and providing a floating object to grab onto.

IX. Appendix

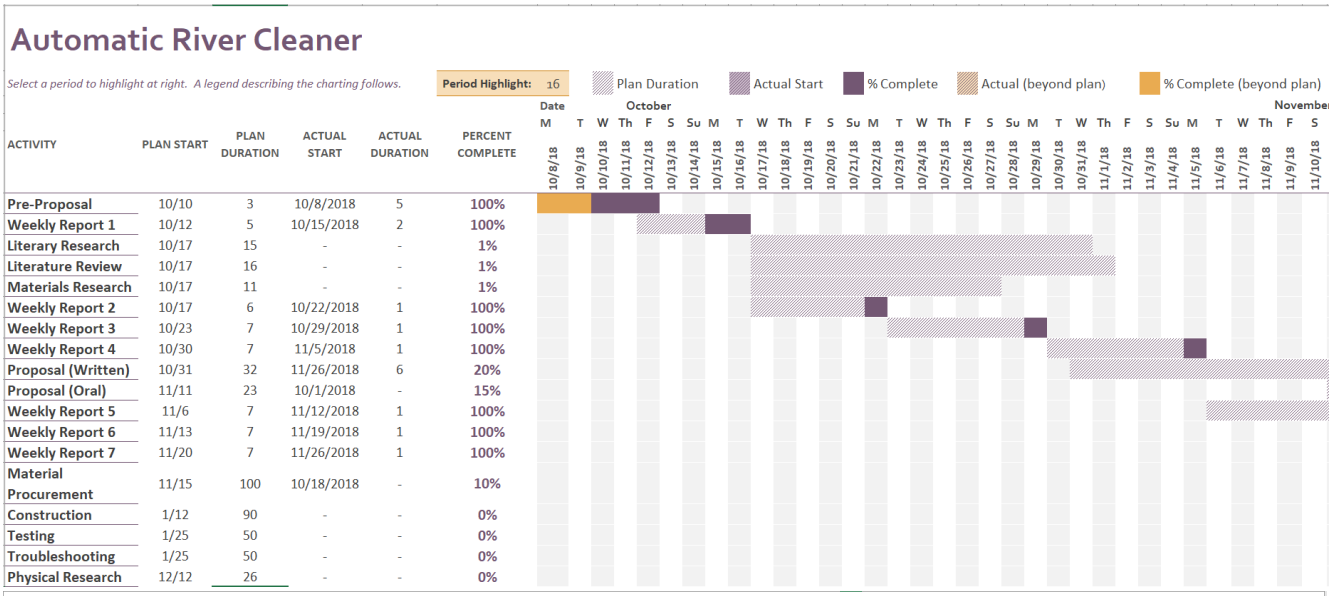
Appendix A.1: Needs and Specifications of the Stakeholder

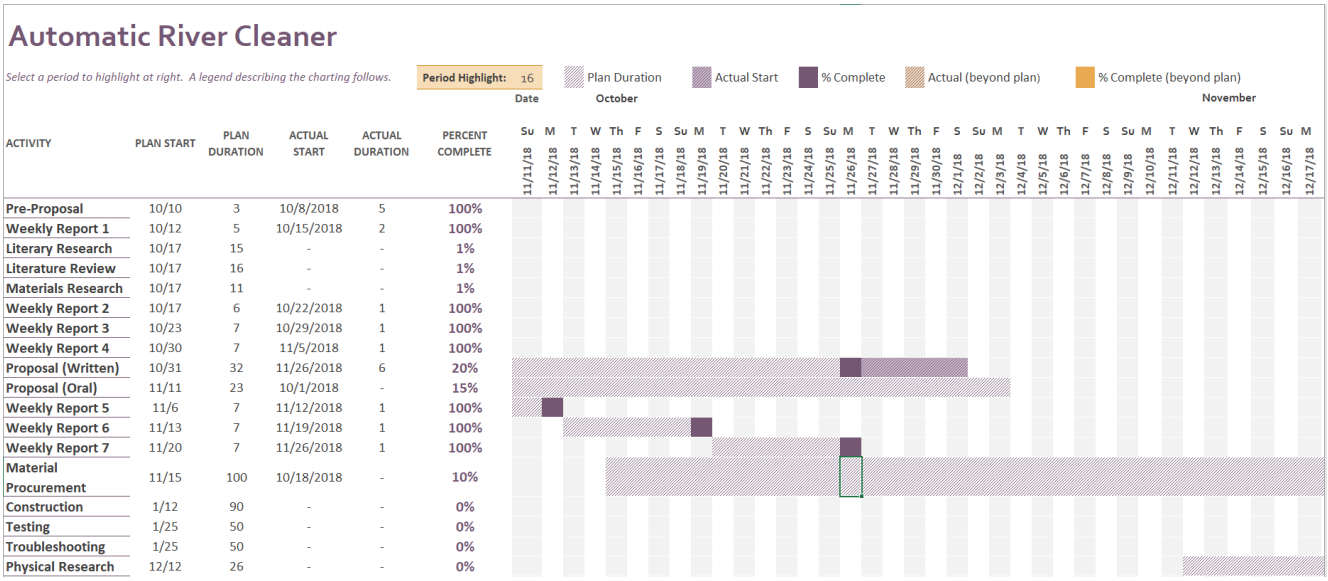
Stakeholder's Needs and Specifications	Ranking
Able to detect ships, living animals, and avoid collision.	10
Robot is reliable and requires minimum daily maintenance.	9
Robot is easy to control and doesn't require more than one person to operate.	6

Appendix A.2: Updated Needs and Specifications of the Stakeholder

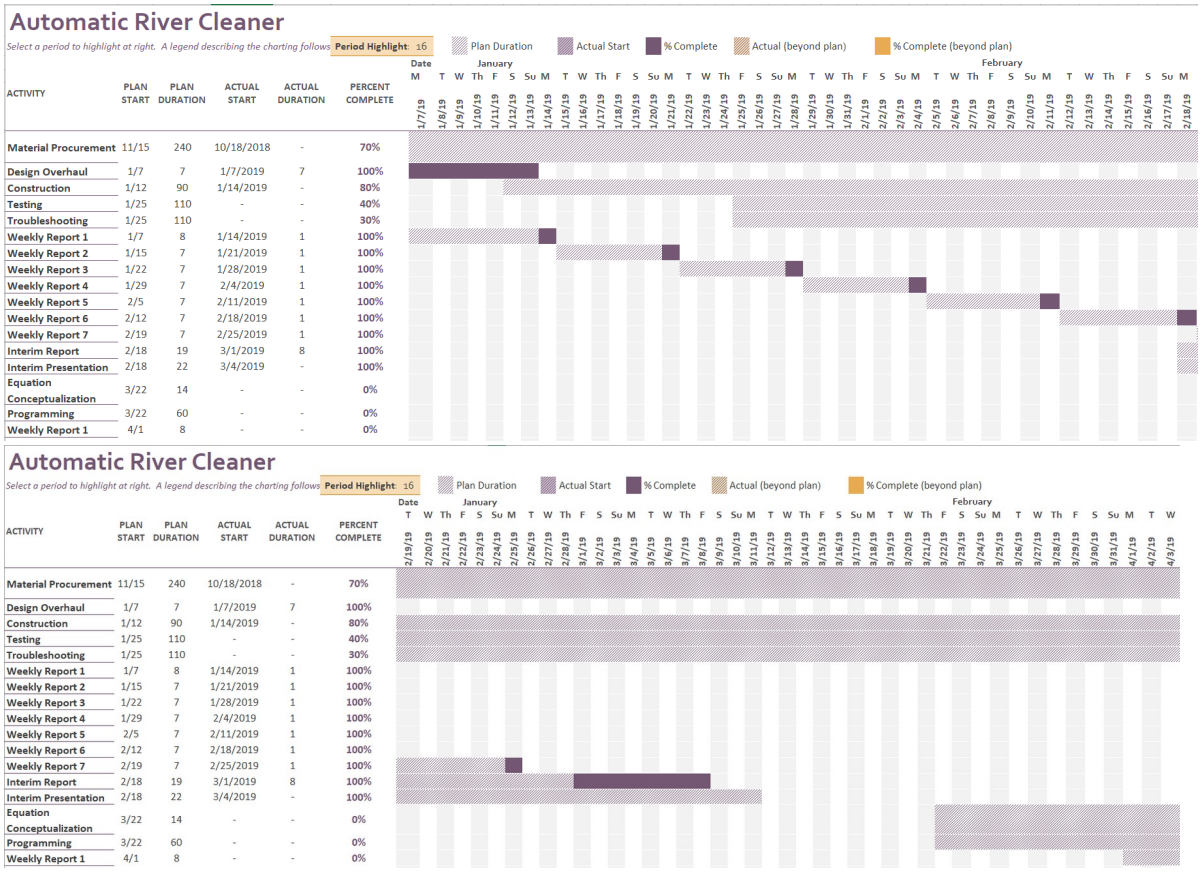
Rank	Needs	Specifications
10	Need the robot to collect the trash efficiently.	Should be able to pick up the trash at least 90% of the time.
9	Need the robot to run economically.	The energy system can run 5 hours without charging
9	Needs, at most, one person to control.	Only needs one man to control the robot to collect the trash.
8	Need to be cost effective.	Cost of the product needs to be lower than $1000.
5	The robot can cruise at a proper speed.	Speed can be remaining at 1 to 2m/s.
10	The robot is reliable.	Robot has a strong structure, and can pass the water test.
7	Operator can control the robot with a long range.	Controlling person can control the robot in, at least, a 500 meter radius.
9	Robot can collect trash fast.	Robot can collect the trash (20 bottles) in 5 minutes.
7	The robot has an object avoidence system	Must be able to avoid collisions with boats and living animals.

Appendix B.1: Proposed Project Schedule For Fall Term

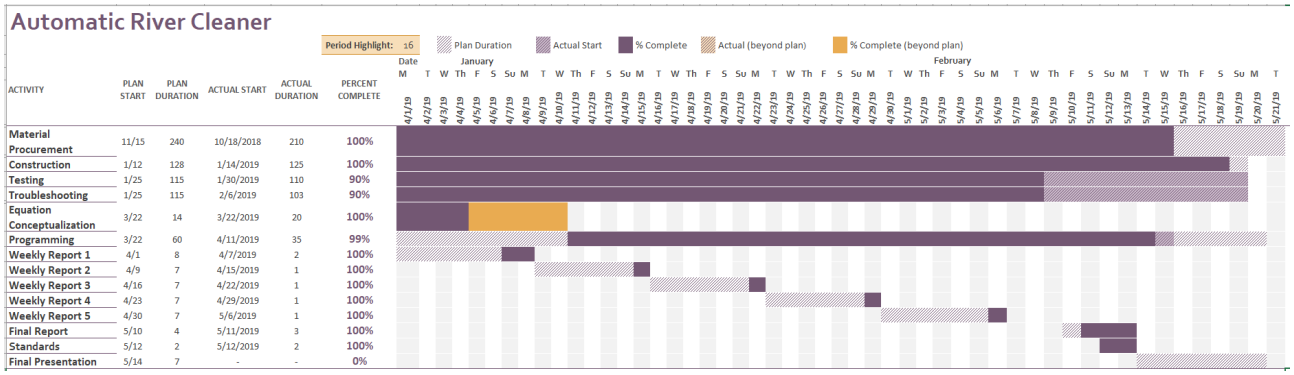




Appendix B.2: Project Schedule For The Winter Term



Appendix B.3: Project Schedule For The Spring Term



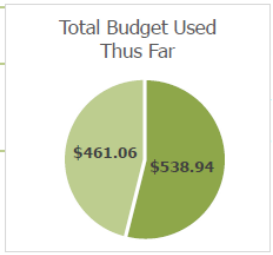
Appendix C.1: Budget List and Cost Breakdown As of The Winter Term

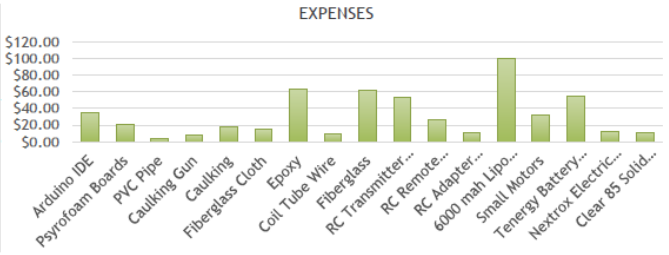
RIVER CLEANER BUDGET

BUDGETED AMOUNT
$1,000.00

TOTAL COSTS
$538.94

DIFFERENCE
$461.06



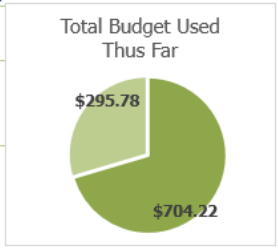


TYPE	MATERIALS/EQUIPMENT	DESCRIPTION	QUANTITY	COST	TOTAL
Electrical	Arduino IDE	Acts as brain for the auto-pilot movement system.	1	$34.89	$34.89
Mechanical	Psyrofoam Boards	High-density, meant for flotation device's walls.	2	$10.50	$21.00
Mechanical	PVC Pipe	1/2" inner diameter	3	$1.50	$4.50
Other	Caulking Gun		1	$8.72	$8.72
Other	Caulking	Will be used as an adhesive for most boat parts.	2	$8.99	$17.98
Mechanical	Fiberglass Cloth	2" width, used for protecting the flotation devices from water degradation.	1	$14.80	$14.80
Other	Epoxy	2 1 gallon containers, connects the fiberglass cloth to the flotation device.	1	$62.97	$62.97
Mechanical	Coil Tube Wire		1	$9.73	$9.73
Mechanical	Fiberglass	27" X 9', same function as the fiberglass cloth	4	$15.45	$61.80
Electrical	RC Transmitter and Receiver	Establishes commuication between user and robot	1	$53.99	$53.99
Electrical	RC Remote Control Receiver	receives signals from user	2	$13.50	$27.00
Electrical	RC Adapter Connector Kit	Supplies female and male connector parts	1	$10.99	$10.99
Electrical	6000 mah Lipo Battery	Used as a power source for boat motors	2	$49.99	$99.98
Mechanical	Small Motors	Source of boat's movement	4	$8.00	$32.00
Electrical	Tenergy Battery Pack Charger	Charges motor batteries	1	$54.99	$54.99
Mechanical	Nextrox Electric Motor	A high torque gear box essential for the trash collector system's movement and operation	1	$11.99	$11.99
Mechanical	Clear 85 Solid Round Belting	Used to transport the torque from the motor to the trash collector system	1	$11.61	$11.61
TOTAL PLANTS					**$538.94**

Appendix C.2: Budget List and Cost Breakdown As of The Spring Term

RIVER CLEANER BUDGET

BUDGETED AMOUNT	TOTAL COSTS	DIFFERENCE
$1,000.00	$704.22	$295.78





TYPE	MATERIALS/EQUIPMENT	DESCRIPTION	QUANTITY	COST	TOTAL
Electrical	Arduino IDE	Acts as brain for the auto-pilot movement system	1	$34.89	$34.89
Mechanical	Psyrofoam Boards	High-density, meant for flotation device's walls.	2	$10.50	$21.00
Mechanical	PVC Pipe	1/2" inner diameter	3	$1.50	$4.50
Other	Caulking Gun		1	$8.72	$8.72
Other	Caulking	Will be used as an adhesive for most boat	2	$8.99	$17.98
Mechanical	Fiberglass Cloth	2" width, used for protecting the flotation devices from water degradation.	1	$14.80	$14.80
Other	Epoxy	2 1 gallon containers, connects the fiberglass cloth to the flotation device.	1	$62.97	$62.97
Mechanical	Coil Tube Wire		1	$9.73	$9.73
Electrical	Black Wire		1	$11.89	$11.89
Electrical	Red Wire		1	$11.97	$11.97
Mechanical	Fiberglass	27" X 9', same function as the fiberglass cloth	4	$15.45	$61.80
Electrical	RC Transmitter and Receiver	Establishes commuication between user and robot	1	$53.99	$53.99
Electrical	RC Remote Control Receiver	receives signals from user	2	$13.50	$27.00
Electrical	RC Adapter Connector Kit	Supplies female and male connector parts	1	$10.99	$10.99
Electrical	6000 mAh Lipo Battery	Used as a power source for boat motors	2	$49.99	$99.98
Mechanical	Small Brushless Motors	Source of boat's movement	4	$8.00	$32.00
Electrical	Tenergy Battery Pack Charger	Charges motor batteries	1	$54.99	$54.99
Mechanical	Nextrox Electric Motor	A high torque gear box essential for the trash collector system's movement and operation	1	$11.99	$11.99
Electrical	Raspberry Pi B+ Motherboard		1	$37.83	$37.83
Electrical	RC Bullet Connector		1	$6.98	$6.98
Electrical	Noir Camera Module		1	$32.19	$32.19
Electrical	8mhz Board Module	for Arduino	1	$6.99	$6.99
Electrical	Joystick Module for Arduino		1	$11.98	$11.98
Electrical	30W 2V-24V Universal		1	$13.99	$13.99
Electrical	32GB EVO Memory Card	Adapter included	1	$7.99	$7.99
Electrical	15W DC-DC Adapter Cords		1	$7.99	$7.99
Electrical	Wireless Bluetooth	Arduino based remote control component	1	$7.49	$7.49
Electrical	Wireless Serial Port Module	Remote Control Component	1	$7.99	$7.99
Mechanical	Clear 85 Solid Round Belting	Used to transport the torque from the motor to the trash collector system	1	$11.61	$11.61
TOTAL COST					**$704.22**

Appendix D.1: Transmitter coding (Sprint Term)

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>
#include <Wire.h>

// Define the digital inputs
#define jB1 1  // Joystick button 1
#define jB2 0  // Joystick button 2
#define t1 7   // Toggle switch 1
#define t2 4   // Toggle switch 1
#define b1 8   // Button 1
#define b2 9   // Button 2
#define b3 2   // Button 3
#define b4 3   // Button 4

const int MPU = 0x68; // MPU6050 I2C address
float AccX, AccY, AccZ;
float GyroX, GyroY, GyroZ;
float accAngleX, accAngleY, gyroAngleX, gyroAngleY;
float angleX, angleY;
float AccErrorX, AccErrorY, GyroErrorX, GyroErrorY;
float elapsedTime, currentTime, previousTime;
int c = 0;


RF24 radio(5, 6);   // nRF24L01 (CE, CSN)
const byte address[6] = "00001"; // Address

// Max size of this struct is 32 bytes - NRF24L01 buffer limit
struct Data_Package {
  byte j1PotX;
  byte j1PotY;
  byte j1Button;
  byte j2PotX;
  byte j2PotY;
  byte j2Button;
  byte pot1;
  byte pot2;
  byte tSwitch1;
  byte tSwitch2;
  byte button1;
  byte button2;
  byte button3;
  byte button4;
};

Data_Package data; //Create a variable with the above structure

void setup() {
  Serial.begin(9600);

  // Initialize interface to the MPU6050
  initialize_MPU6050();

  // Call this function if you need to get the IMU error values for your module
  //calculate_IMU_error();

  // Define the radio communication
  radio.begin();
  radio.openWritingPipe(address);
  radio.setAutoAck(false);
  radio.setDataRate(RF24_250KBPS);
  radio.setPALevel(RF24_PA_LOW);

  // Activate the Arduino internal pull-up resistors
  pinMode(jB1, INPUT_PULLUP);
  pinMode(jB2, INPUT_PULLUP);
  pinMode(t1, INPUT_PULLUP);
  pinMode(t2, INPUT_PULLUP);
  pinMode(b1, INPUT_PULLUP);
  pinMode(b2, INPUT_PULLUP);
  pinMode(b3, INPUT_PULLUP);
  pinMode(b4, INPUT_PULLUP);

  // Set initial default values
  data.j1PotX = 127; // Values from 0 to 255.
  data.j1PotY = 127;
  data.j2PotX = 127;
  data.j2PotY = 127;
```

```
  data.j1Button = 1;
  data.j2Button = 1;
  data.pot1 = 1;
  data.pot2 = 1;
  data.tSwitch1 = 1;
  data.tSwitch2 = 1;
  data.button1 = 1;
  data.button2 = 1;
  data.button3 = 1;
  data.button4 = 1;
}
void loop() {
  // Read all analog inputs and map them to one Byte value
  data.j1PotX = map(analogRead(A1), 0, 1023, 0, 255); // Convert the analog read value into a BYTE value from 0 to 255
  data.j1PotY = map(analogRead(A0), 0, 1023, 0, 255);
  data.j2PotX = map(analogRead(A2), 0, 1023, 0, 255);
  data.j2PotY = map(analogRead(A3), 0, 1023, 0, 255);
  data.pot1 = map(analogRead(A7), 0, 1023, 0, 255);
  data.pot2 = map(analogRead(A6), 0, 1023, 0, 255);
  // Read all digital inputs
  data.j1Button = digitalRead(jB1);
  data.j2Button = digitalRead(jB2);
  data.tSwitch2 = digitalRead(t2);
  data.button1 = digitalRead(b1);
  data.button2 = digitalRead(b2);
  data.button3 = digitalRead(b3);
  data.button4 = digitalRead(b4);
  // If toggle switch 1 is switched on
  if (digitalRead(t1) == 0) {
    read_IMU();    // Use MPU6050 instead of Joystick 1 for controling left, right, forward and backward movements
  }
  // Send the whole data from the structure to the receiver
  radio.write(&data, sizeof(Data_Package));
}

void initialize_MPU6050() {
  Wire.begin();                      // Initialize comunication
  Wire.beginTransmission(MPU);       // Start communication with MPU6050 // MPU=0x68
  Wire.write(0x6B);                  // Talk to the register 6B
  Wire.write(0x00);                  // Make reset - place a 0 into the 6B register
  Wire.endTransmission(true);        //end the transmission
  // Configure Accelerometer
  Wire.beginTransmission(MPU);
  Wire.write(0x1C);                  //Talk to the ACCEL_CONFIG register
  Wire.write(0x10);                  //Set the register bits as 00010000 (+/- 8g full scale range)
  Wire.endTransmission(true);
  // Configure Gyro
  Wire.beginTransmission(MPU);
  Wire.write(0x1B);                   // Talk to the GYRO_CONFIG register (1B hex)
  Wire.write(0x10);                   // Set the register bits as 00010000 (1000dps full scale)
  Wire.endTransmission(true);
}

void calculate_IMU_error() {
  // Read accelerometer values 200 times
  while (c < 200) {
    Wire.beginTransmission(MPU);
    Wire.write(0x3B);
    Wire.endTransmission(false);
    Wire.requestFrom(MPU, 6, true);
    AccX = (Wire.read() << 8 | Wire.read()) / 4096.0 ;
    AccY = (Wire.read() << 8 | Wire.read()) / 4096.0 ;
    AccZ = (Wire.read() << 8 | Wire.read()) / 4096.0 ;
    // Sum all readings
    AccErrorX = AccErrorX + ((atan((AccY) / sqrt(pow((AccX), 2) + pow((AccZ), 2))) * 180 / PI));
    AccErrorY = AccErrorY + ((atan(-1 * (AccX) / sqrt(pow((AccY), 2) + pow((AccZ), 2))) * 180 / PI));
    c++;
  }
  //Divide the sum by 200 to get the error value
  AccErrorX = AccErrorX / 200;
  AccErrorY = AccErrorY / 200;
  c = 0;
  // Read gyro values 200 times
  while (c < 200) {
    Wire.beginTransmission(MPU);
    Wire.write(0x43);
    Wire.endTransmission(false);
    Wire.requestFrom(MPU, 4, true);
    GyroX = Wire.read() << 8 | Wire.read();
    GyroY = Wire.read() << 8 | Wire.read();
    // Sum all readings
    GyroErrorX = GyroErrorX + (GyroX / 32.8);
```

```
    GyroErrorY = GyroErrorY + (GyroY / 32.8);
    c++;
  }
  //Divide the sum by 200 to get the error value
  GyroErrorX = GyroErrorX / 200;
  GyroErrorY = GyroErrorY / 200;
  // Print the error values on the Serial Monitor
  Serial.print("AccErrorX: ");
  Serial.println(AccErrorX);
  Serial.print("AccErrorY: ");
  Serial.println(AccErrorY);
  Serial.print("GyroErrorX: ");
  Serial.println(GyroErrorX);
  Serial.print("GyroErrorY: ");
  Serial.println(GyroErrorY);
}

void read_IMU() {
  // === Read acceleromter data === //
  Wire.beginTransmission(MPU);
  Wire.write(0x3B); // Start with register 0x3B (ACCEL_XOUT_H)
  Wire.endTransmission(false);
  Wire.requestFrom(MPU, 6, true); // Read 6 registers total, each axis value is stored in 2 registers
  //For a range of +-8g, we need to divide the raw values by 4096, according to the datasheet
  AccX = (Wire.read() << 8 | Wire.read()) / 4096.0; // X-axis value
  AccY = (Wire.read() << 8 | Wire.read()) / 4096.0; // Y-axis value
  AccZ = (Wire.read() << 8 | Wire.read()) / 4096.0; // Z-axis value

  // Calculating angle values using
  accAngleX = (atan(AccY / sqrt(pow(AccX, 2) + pow(AccZ, 2))) * 180 / PI) + 1.15; // AccErrorX ~(-1.15)
  accAngleY = (atan(-1 * AccX / sqrt(pow(AccY, 2) + pow(AccZ, 2))) * 180 / PI) - 0.52; // AccErrorX ~(0.5)

  // === Read gyro data === //
  previousTime = currentTime;        // Previous time is stored before the actual time read
  currentTime = millis();            // Current time actual time read
  elapsedTime = (currentTime - previousTime) / 1000;   // Divide by 1000 to get seconds
  Wire.beginTransmission(MPU);
  Wire.write(0x43); // Gyro data first register address 0x43
  Wire.endTransmission(false);
  Wire.requestFrom(MPU, 4, true); // Read 4 registers total, each axis value is stored in 2 registers
  GyroX = (Wire.read() << 8 | Wire.read()) / 32.8; // For a 1000dps range we have to divide first the raw value by 32.8, according to the datasheet
  GyroY = (Wire.read() << 8 | Wire.read()) / 32.8;
  GyroX = GyroX + 1.85; //// GyroErrorX ~(-1.85)
  GyroY = GyroY - 0.15; // GyroErrorY ~(0.15)
  // Currently the raw values are in degrees per seconds, deg/s, so we need to multiply by sendonds (s) to get the angle in degrees
  gyroAngleX = GyroX * elapsedTime;
  gyroAngleY = GyroY * elapsedTime;

  // Complementary filter - combine acceleromter and gyro angle values
  angleX = 0.98 * (angleX + gyroAngleX) + 0.02 * accAngleX;
  angleY = 0.98 * (angleY + gyroAngleY) + 0.02 * accAngleY;
  // Map the angle values from -90deg to +90 deg into values from 0 to 255, like the values we are getting from the Joystick
  data.j1PotX = map(angleX, -90, +90, 255, 0);
  data.j1PotY = map(angleY, -90, +90, 0, 255);
}
```

Appendix D.2: Receiver coding (Sprint Term)

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>
#include <Servo.h>

Servo ESC1;     // create servo object to control the ESC
Servo ESC2;     // create servo object to control the ESC
int potValue;  // value from the analog pin
int X;
int Y;
int diff;
int rspeed;
int lspeed;
int rmax;
int lmax;
int level;


RF24 radio(53, 48);   // nRF24L01 (CE, CSN)
const byte address[6] = "00001";

unsigned long lastReceiveTime = 0;
unsigned long currentTime = 0;

// Max size of this struct is 32 bytes - NRF24L01 buffer limit
struct Data_Package {
  byte j1PotX;
  byte j1PotY;
  byte j1Button;
  byte j2PotX;
  byte j2PotY;
  byte j2Button;
  byte pot1;
  byte pot2;
  byte tSwitch1;
  byte tSwitch2;
  byte button1;
  byte button2;
  byte button3;
  byte button4;
};

Data_Package data; //Create a variable with the above structure

void setup() {
  ESC1.attach(9,800,1100); // (pin, min pulse width, max pulse width in microseconds)
  ESC2.attach(10,800,1100);
  Serial.begin(9600);
  radio.begin();
  radio.openReadingPipe(0, address);
  radio.setAutoAck(false);
  radio.setDataRate(RF24_250KBPS);
  radio.setPALevel(RF24_PA_LOW);
  radio.startListening(); //  Set the module as receiver
  resetData();
}
void loop() {
  // Check whether there is data to be received
  if (radio.available()) {
    radio.read(&data, sizeof(Data_Package)); // Read the whole data and store it into the 'data' structure
    lastReceiveTime = millis(); // At this moment we have received the data
  }
  // Check whether we keep receving data, or we have a connection between the two modules
  currentTime = millis();
  if ( currentTime - lastReceiveTime > 1000 ) {
    // If current time is more then 1 second since the last data, that means we have lost connection
    resetData(); // If connection is lost, reset the data.
  }

  // // Print the data in the Serial Monitor
  // Serial.print("j1PotX: ");
  // Serial.print(data.j1PotX);
  // Serial.print("; j1PotY: ");
  // Serial.print(data.j1PotY);
  // Serial.print("; button1: ");
  // Serial.print(data.button1);

  potValue = data.pot1;// scale it to use it with the servo library (value between 0 and 255)
  X = data.j1PotX;

```

```
  if ( data.j1PotY < 127 ){
    ESC1.write(0); //right
    ESC2.write(0); //left

  }else{
    Y = map(data.j1PotY, 127, 255, 0, 127);
    if (data.j1PotX < 127){
      X = map(data.j1PotX, 127, 0, 0, 127);
      diff = Y-X;
      diff = map(diff, -127, 127 , 0, 255);
      level = max(X,Y);
      level = map(level, 0, 127, 0, 255);
      rmax = potValue * level;
      lmax = potValue * diff;
      rspeed = map(rmax, 0, 32385, 0 ,180);
      lspeed = map(lmax, 0, 65025, 0, 180);
      ESC1.write(rspeed); //right
      ESC2.write(lspeed);
    else if ( data.j1PotX > 127) {
      X = map(data.j1PotX, 127, 255, 0, 127);
      diff = Y-X;
      diff = map(diff, -127, 127 , 0, 255);
      level = max(X,Y);
      level = map(level, 0, 127, 0, 255);
      rmax = potValue * diff;
      lmax = potValue * level;
      rspeed = map(rmax, 0, 65025, 0 ,180);
      lspeed = map(lmax, 0, 32385, 0, 180);
      ESC1.write(rspeed); //right
      ESC2.write(lspeed);
    }
    else {
      rmax = potValue * Y;
      lmax = potValue * Y;
      rspeed = map(rmax, 0, 32385, 0 ,180);
      lspeed = map(lmax, 0, 32385, 0, 180);
      ESC1.write(rspeed); //right
      ESC2.write(lspeed);
    }
    }

  }


}

void resetData() {
  // Reset the values when there is no radio connection - Set initial default values
  data.j1PotX = 127;
  data.j1PotY = 127;
  data.j2PotX = 127;
  data.j2PotY = 127;
  data.j1Button = 1;
  data.j2Button = 1;
  data.pot1 = 1;
  data.pot2 = 1;
  data.tSwitch1 = 1;
  data.tSwitch2 = 1;
  data.button1 = 1;
  data.button2 = 1;
  data.button3 = 1;
  data.button4 = 1;
}
```

X. Reference

[1] Kennedy, Kimberly T. M., and Rana W. El-Sabaawi. “Decay Patterns of Invasive Plants and Plastic Trash in Urban Streams.” *Urban Ecosystems*, vol. 21, no. 5, 2018, pp. 817–830., doi:10.1007/s11252-018-0771-9.

[2] Zhang, Kai, et al. “Accumulation of Floating Microplastics behind the Three Gorges Dam.” *Environmental Pollution*, vol. 204, 2015, pp. 117–123., doi:10.1016/j.envpol.2015.04.023.

[3] Xu, Pei, et al. “Microplastic Risk Assessment in Surface Waters: A Case Study in the Changjiang Estuary, China.” *Marine Pollution Bulletin*, vol. 133, 2018, pp. 647–654., doi:10.1016/j.marpolbul.2018.06.020.

[4] “Yangtze River Facts.” *Table Manners in China, Etiquettes and Taboos When Dining*, www.travelchinaguide.com/river/yangtze/

[5] Thompson, Avery. “You Could Drive This River-Cleaning Trashbot From Anywhere.” Popular Mechanics，Popular Mechanics, 3 Apr. 2018, www.popularmechanics.com/technology/robots/a19676406/chicago-river-trashbot-urban-rivers/

[6] Jackie, Snow. “Googly-Eyed Trash Eaters May Clean a Harbor Near You.” National Geographic, National Geographic Society, 3 Mar. 2017, http://news.nationalgeographic.com/2017/02/mr-trash-wheels-professor-trash-wheels-baltimore-harbor-ocean-trash-pickup/